

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 1: Subsonic Airflow, Continuity & Bernoulli's Principle

Syllabus Source: CAE Oxford (pp. 1-34) | Pass Target: >= 90% AviationExam

1. AIR PROPERTIES, ATMOSPHERE & THE SPEED OF SOUND

Aerodynamics investigates the interaction between moving air and solid airframe bodies. Air behaves as an ideal compressible gas:

- Air Density (ρ): Mass per unit volume (kg/m^3). Standard sea level density (ISA ρ_0) = 1.225 kg/m^3 . Density decreases with increasing altitude, increasing temperature, and increasing relative humidity (humid air is LESS DENSE than dry air because H_2O molecular weight [18] is less than N_2 [28] and O_2 [32])!
- Static Pressure (p): The weight of the atmospheric column acting equally in all directions. Standard MSL pressure = $1013.25 \text{ hPa} = 101,325 \text{ N/m}^2$.
- Speed of Sound (a): The propagation velocity of pressure waves: $a = \sqrt{\gamma \times R \times T} = 38.94 \times \sqrt{T \text{ [Kelvin]}}$. The speed of sound depends EXCLUSIVELY on absolute temperature! At standard ISA sea level ($+15^\circ\text{C} / 288 \text{ K}$): $a = 661.5 \text{ kt} = 340 \text{ m/s}$.

2. THE CONTINUITY EQUATION & STREAMTUBE FLOW

For steady, continuous fluid flow through a streamtube, the principle of conservation of mass dictates that mass flow rate ($\dot{m}$) is constant:

- Continuity Equation: Mass Flow Rate: $\dot{m} = \rho \times A \times V = \text{Constant}$ (where A = cross-sectional area, V = flow velocity).
- Incompressible Flow (Subsonic Mach < 0.3): Air density is assumed constant ($\rho = \text{constant}$). Therefore: Area $\times$ Velocity = Constant: $A_1 \times V_1 = A_2 \times V_2$. As a streamtube CONVERGES (area decreases), velocity must INCREASE. As it DIVERGES (area expands), velocity must DECREASE.

3. BERNOULLI'S THEOREM & VENTURI EFFECT

Bernoulli's theorem represents the conservation of energy in steady, incompressible, frictionless streamline flow:

- Total Pressure (Stagnation Pressure): Total Pressure (P_t) = Static Pressure (P_s) + Dynamic Pressure (q) = Constant.
- Dynamic Pressure Formula: $q = \frac{1}{2} \rho V^2$. Dynamic pressure represents the kinetic energy of the moving fluid.
- Venturi Throat Dynamics: As air flows into a convergent throat: Area narrows $\rightarrow$ Velocity increases $\rightarrow$ Dynamic pressure increases $\rightarrow$ Static pressure DROPS! Downstream divergent section: Area expands $\rightarrow$ Velocity decreases $\rightarrow$ Static pressure RISES.

Regime / Duct Section	Cross-Sectional Area (A)	Velocity (V)	Dynamic Pressure (q)	Static Pressure (P_s)
Subsonic Convergent Duct	Decreases	INCREASES	INCREASES	DECREASES (Bernoulli drop)
Subsonic Divergent Duct	Increases	DECREASES	DECREASES	INCREASES (Diffuser action)
Supersonic Convergent Duct	Decreases	DECREASES	DECREASES	INCREASES (Compressibility dominates)

Supersonic Divergent Duct	Increases	INCREASES	INCREASES	DECREASES (Nozzle expansion)
----------------------------------	-----------	-----------	-----------	------------------------------

[!] EXAM TRAP (AVIATIONEXAM) - Air Density vs Humidity Exam Trap

- QUESTION: Does high relative humidity increase or decrease air density? - ANSWER: High humidity DECREASES air density! - Water vapor (H₂O) has a molecular weight of 18 g/mol, whereas dry air (N₂ and O₂) averages 29 g/mol. Humid air is lighter and less dense than dry air at the same temperature and pressure, reducing aerodynamic lift and engine thrust!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Speed of Sound & Humidity**

Step 1 (Speed of Sound): To find local acoustic speed 'a' at any flight level, calculate OAT in Kelvin ($K = C + 273$). Apply $a = 38.94 \times \sqrt{T_K}$. Example: at -56.5 C (216.5 K), $a = 38.94 \times \sqrt{216.5} = 573$ kt. Step 2 (Humidity Trap): Humid air is LESS DENSE than dry air. Water vapor (H₂O = 18 g/mol) is lighter than nitrogen (28) and oxygen (32). High temperature and high humidity reduce density, requiring longer takeoff runs.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: According to Bernoulli's equation for steady incompressible flow, what happens to static pressure when a

- [A]: Static pressure increases
- [B]: Static pressure DECREASES while dynamic pressure increases, keeping total pressure constant
- [C]: Total pressure drops to zero
- [D]: Air density increases

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. By Bernoulli's principle, Total Pressure = Static + Dynamic. When velocity accelerates, dynamic pressure ($\frac{1}{2} \rho V^2$) rises; static pressure must decrease by an equal amount to conserve energy.

Q2: On what single atmospheric variable does the local speed of sound in air depend?

- [A]: Static pressure only
- [B]: Air density only
- [C]: Absolute Air Temperature (Kelvin) ONLY
- [D]: Relative humidity

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Formula: $a = \sqrt{\gamma \times R \times T} = 38.94 \times \sqrt{T}$. The speed of sound depends exclusively on absolute air temperature in Kelvin. Pressure and density changes cancel each other out.

Q3: How does high relative atmospheric humidity affect aircraft aerodynamic take-off performance?

- [A]: Increases lift and improves engine power

- [B]: Reduces air density, which reduces aerodynamic lift and degrades engine take-off thrust
- [C]: Increases the speed of sound
- [D]: Has zero effect on aviation

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Water molecules are lighter than nitrogen and oxygen molecules. Humid air reduces air density, decreasing dynamic pressure for a given speed, lengthening take-off ground run.

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 2: Airfoil Geometry, Lift Generation & Pitching Moments

Syllabus Source: CAE Oxford (pp. 35-72) | Pass Target: >= 90% AviationExam

1. AIRFOIL ANATOMY & GEOMETRIC TERMINOLOGY

An airfoil is an aerodynamic cross-section engineered to generate usable lift while minimizing drag:

- **Chord Line:** A straight reference line connecting the leading edge and trailing edge centers of curvature. Chord length (c) is the reference datum for aerodynamic dimensions.
- **Mean Camber Line (MCL):** A line equidistant from the upper and lower airfoil surfaces. If the MCL coincides with the chord line, the airfoil is **SYMMETRICAL**. If MCL lies above chord, it is **POSITIVELY CAMBERED**.
- **Maximum Camber:** The maximum perpendicular distance between chord line and MCL, expressed as a percentage of chord (typically 2% to 4% at 30% to 40% chord). Camber determines zero-lift angle of attack.
- **Maximum Thickness:** The maximum distance between upper and lower surfaces (t/c ratio, e.g. 12% to 15% for transport aircraft, 9% to 11% for high-speed transonic airfoils).
- **Angle of Attack (Alpha):** The acute angle between the airfoil chord line and the **RELATIVE AIRFLOW** (freestream vector). Alpha is completely independent of the aircraft's pitch attitude relative to the horizon!

2. LIFT GENERATION: CIRCULATION, DOWNWASH & PRESSURE DISTRIBUTION

Lift is generated through two complementary aerodynamic mechanisms:

- **Pressure Differential (Bernoulli):** Air flowing over the curved upper surface accelerates, causing static pressure to drop below ambient (creating suction / low pressure). Air on the lower surface experiences slight compression. Upper surface suction contributes 70% to 80% of total aerodynamic lift!
- **Newton's Third Law (Downwash Deflection):** The airfoil deflects oncoming air downwards (downwash angle). The reaction to accelerating air downwards is an equal and opposite upward force: $LIFT = Downward\ Mass\ Flow \times Vertical\ Acceleration$.

3. CENTER OF PRESSURE (CP) VS AERODYNAMIC CENTER (AC)

Understanding the distinction between Center of Pressure and Aerodynamic Center is critical for aircraft stability:

Aerodynamic Datum	Physical Definition & Location	Behavior with Angle of Attack Changes
Center of Pressure (CP)	Point along chord where total resultant aerodynamic force acts.	MOVES with Angle of Attack! On a positively cambered airfoil: As AOA increases, CP MOVES FORWARD toward leading edge (up to stall). As AOA decreases, CP moves REARWARD. Inherently unstable!
Aerodynamic Center (AC)	Point along chord about which the pitching moment coefficient remains CONSTANT regardless of AOA.	DOES NOT MOVE with AOA in subsonic flight! Located at approximately the 25% CHORD POINT (Quarter-Chord) for all subsonic airfoils (moves to 50% chord in supersonic flight).

[!] EXAM TRAP (AVIATIONEXAM) - Pitching Moment About the Aerodynamic Center

- Symmetrical Airfoil: Pitching moment about the AC is ZERO at all angles of attack!
 - Positively Cambered Airfoil: Generates a NOSE-DOWN (negative) pitching moment about the AC that remains CONSTANT as AOA changes!
 - Center of Pressure Movement: On cambered wings, increasing AOA moves CP FORWARD; on symmetrical wings, CP remains virtually fixed at 25% chord.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): CP vs AC Identification**

Rule 1 (Aerodynamic Centre): If the question mentions 'constant pitching moment coefficient independent of angle of attack' or '25% chord subsonic', the answer is ALWAYS the Aerodynamic Centre (AC).
 Rule 2 (Centre of Pressure): If the question mentions 'point where total aerodynamic lift acts' or 'moves forward with increasing angle of attack on cambered airfoils', the answer is the Centre of Pressure (CP).
 Rule 3 (Symmetric Airfoil): On a purely symmetrical airfoil, CP and AC coincide at 25% chord and CP does not move within normal AoA.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: For a positively cambered subsonic airfoil, how does the Center of Pressure (CP) move when the Angle of Attack (AOA) increases?**

- [A]: CP moves rearward toward the trailing edge
- [B]: CP moves FORWARD toward the leading edge
- [C]: CP remains fixed at 50% chord
- [D]: CP moves downward

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. On positively cambered airfoils, increasing AOA intensifies leading-edge suction peak, drawing the resultant Center of Pressure forward toward the quarter-chord point.

Q2: What is the defining characteristic of an airfoil's Aerodynamic Center (AC) in subsonic flight?

- [A]: Lift is always zero at the AC
- [B]: The pitching moment about the AC remains CONSTANT as angle of attack changes, located at ~25% chord
- [C]: Total drag is zero at the AC
- [D]: It moves forward with speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Aerodynamic Center is the unique theoretical point where pitching moment coefficient is independent of AOA. In subsonic flow, it is fixed at approximately 25% chord.

Q3: What contributes the greatest percentage of total aerodynamic lift generated by an aircraft wing?

- [A]: High pressure pushing up against the lower surface (20-30%)
- [B]: Low static pressure (suction) acting on the UPPER surface (70-80%)
- [C]: Engine exhaust thrust

- [D]: Propeller wash

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Upper surface low pressure (suction) accounts for approximately 70% to 80% of total lift; lower surface positive pressure contributes only 20% to 30%.

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 3: Lift & Drag Coefficients, Formulas & Polar Curves

Syllabus Source: CAE Oxford (pp. 73-108) | Pass Target: >= 90% AviationExam

1. THE AERODYNAMIC LIFT FORMULA

Total aerodynamic lift generated by a wing is governed by the universal lift formula:

- Lift Equation: $L = C_L \times \frac{1}{2} \rho V^2 \times S$, where C_L = Lift Coefficient (dimensionless); ρ = ambient air density (kg/m^3); V = True Airspeed (TAS, m/s); S = Wing surface area (m^2).
- Dynamic Pressure Component ($q = \frac{1}{2} \rho V^2$): Lift is directly proportional to dynamic pressure and wing planform area.
- Lift Coefficient (C_L): Represents airfoil lifting efficiency. Depends on: 1. Angle of attack (α); 2. Camber; 3. Airfoil shape; 4. Reynolds Number; 5. Mach number.
- C_L vs α Curve: Linear increase from zero-lift angle of attack up to $C_{L_{\text{max}}}$. For positively cambered airfoils, zero lift occurs at a NEGATIVE angle of attack (typically -2° to -4°). Symmetrical airfoils produce zero lift at EXACTLY 0° AOA.

2. THE DRAG FORMULA & DRAG COEFFICIENT (C_D)

Total aerodynamic drag resists aircraft forward motion through the air:

- Drag Equation: $D = C_D \times \frac{1}{2} \rho V^2 \times S$, where C_D = Drag Coefficient.
- Total Drag Decomposition: Total Drag = Parasite Drag (Profile Drag) + Induced Drag (Vortex Drag): $C_D = C_{Dp} + C_{Di}$.
- Lift-to-Drag Ratio (L/D): $L/D = C_L / C_D$. Measures aerodynamic efficiency. Maximum L/D ratio (L/D_{max}) occurs at the tangent from the origin to the polar curve, corresponding to the MINIMUM DRAG SPEED (V_{md})!

3. THE DRAG POLAR (LILIENTHAL POLAR CURVE)

Key Polar Feature	Aerodynamic Meaning & Optimum Point	Flight Application & Performance Target
Key Polar Feature	Aerodynamic Meaning & Optimum Point	Flight Application & Performance Target
Origin Tangent Point	Maximum Lift-to-Drag Ratio: (C_L / C_D) _{max} .	Minimum Drag Speed (V_{md}). Yields: 1. Maximum Glide Range (unpowered); 2. Maximum Endurance for jet aircraft; 3. Maximum Range for piston-prop aircraft.
Peak of the Curve	Maximum Lift Coefficient ($C_{L_{\text{max}}}$).	Minimum Flying Airspeed (Stall Speed V_S). Corresponds to stalling angle of attack (α_{crit} , $\sim 15\text{-}18^\circ$).
Maximum ($C_L^{1/2} / C_D$)	Optimum speed for jet aircraft cruising range.	Long-range cruise speed for jet airliners (typically $1.32 \times V_{md}$).
Maximum ($C_L^{3/2} / C_D$)	Minimum rate of sink / Minimum power required.	Maximum endurance for propeller aircraft; minimum sink speed for gliders (typically $0.76 \times V_{md}$).

[!] EXAM TRAP (AVIATIONEXAM) - Glide Performance vs Aircraft Gross Mass Trap

- QUESTION: How does increasing aircraft gross mass affect the maximum glide distance in power-off flight? - ANSWER: Gross mass HAS ZERO EFFECT ON MAXIMUM GLIDE DISTANCE! - Maximum glide ratio equals $(L/D)_{max}$, which is purely aerodynamic. - Effect of Heavier Weight: The aircraft glides at a HIGHER TRUE AIRSPEED and has a HIGHER RATE OF DESCENT, covering the exact same ground distance in a shorter time!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Weight Variation in Gliding Flight**

Problem: An aircraft glides at 100 kt with engines idle. If gross weight increases by 21%, what happens to glide range and best glide speed? Step 1: Glide range depends exclusively on L/D_{max} . Clean aerodynamics are unchanged, so glide range in still air DOES NOT CHANGE. Step 2: Best glide speed increases with the square root of weight ratio: $V_{new} = 100 \times \sqrt{1.21} = 100 \times 1.10 = 110$ kt. Conclusion: The heavier aircraft glides at 110 kt vs 100 kt, reaches the ground in less time, but lands at the EXACT SAME distance.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: If an aircraft doubles its airspeed ($2 \times V$) while maintaining constant altitude and angle of attack, by what factor does lift increase?

- [A]: By a factor of 2
- [B]: By a factor of 4 (quadruples)
- [C]: By a factor of 8
- [D]: Lift remains constant

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Lift formula: $L = C_L \times \frac{1}{2} \rho V^2 \times S$. Because lift is proportional to the square of velocity (V^2), doubling airspeed produces $(2)^2 = 4$ times more lift.

Q2: At what angle of attack does a SYMMETRICAL airfoil generate zero aerodynamic lift?

- [A]: At -4° angle of attack
- [B]: At exactly ZERO degrees (0°) angle of attack
- [C]: At $+2^\circ$ angle of attack
- [D]: At stalling angle

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Symmetrical airfoils have identical upper and lower contours with zero camber. Airflow acceleration is identical on both surfaces at 0° AOA, producing zero net lift. (Positively cambered airfoils require negative AOA for zero lift).

Q3: What operational flight condition corresponds to the point of Maximum Lift-to-Drag Ratio $((L/D)_{max})$?

- [A]: Maximum airspeed
- [B]: Minimum Total Drag Speed (V_{md}) and Maximum Glide Range
- [C]: Maximum stall speed
- [D]: Minimum climb angle

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At $(L/D)_{\max}$, aerodynamic efficiency is optimized. Total drag is at its absolute minimum (V_{md}), providing the shallowest glide angle and maximum unpowered glide distance.

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 4: Three-Dimensional Wing Airflow & Induced Drag

Syllabus Source: CAE Oxford (pp. 109-144) | Pass Target: >= 90% AviationExam

1. THREE-DIMENSIONAL AIRFLOW & WINGTIP VORTICES

A real finite wing produces 3D airflow due to spanwise pressure differentials between upper and lower wing surfaces:

- Spanwise Flow: Air on the high-pressure lower surface flows OUTWARD toward wingtips; air on the low-pressure upper surface flows INWARD toward fuselage. At the trailing edge and wingtips, these opposing spanwise flows roll up into trailing WINGTIP VORTICES.
- Induced Downwash (w): Wingtip vortices trail downstream and induce a strong downward vertical velocity component behind the wing called induced downwash. Downwash tilts the local relative airflow vector downwards!
- Effective Angle of Attack: Effective AOA = Geometric AOA - Induced Angle (α_i). Because the relative airflow is tilted downward, the total aerodynamic lift vector tilts REARWARD. The rearward-tilted component of lift is INDUCED DRAG (Vortex Drag)!

2. INDUCED DRAG FORMULA & WING GEOMETRY FACTORS

Induced drag is the inevitable aerodynamic price of generating lift with a finite-span wing:

- Induced Drag Coefficient: $C_{Di} = (C_L^2) / (\pi \times AR \times e)$, where AR = Aspect Ratio, e = Oswald wing efficiency factor (~0.8 to 0.9).
- Variation with Speed: Because C_L is inversely proportional to V^2 in level flight ($C_L = W / (q S)$), Induced Drag is INVERSELY PROPORTIONAL TO SPEED SQUARED: D_i proportional to $1 / V^2$! Induced drag is massive at low speeds (take-off, climb, landing approach) and becomes negligible at high cruising speeds.
- Variation with Mass: Induced drag is directly proportional to the SQUARE OF AIRCRAFT GROSS MASS: D_i proportional to W^2 ! A 20% increase in mass increases induced drag by $(1.20)^2 = 44\%$!

Design Factor	Aerodynamic Influence on Induced Drag	Aeronautical Examples & Trade-offs
Design Factor	Aerodynamic Influence on Induced Drag	Aeronautical Examples & Trade-offs
Aspect Ratio (AR = b^2 / S)	HIGH Aspect Ratio dramatically REDUCES induced drag (spreads lift over longer wingspan, reducing vortex strength).	Gliders (AR ~ 25 to 30), high-altitude surveillance (U-2). Limited by wing root bending moments and structural weight.
Wing Planform Taper	Elliptical lift distribution produces minimum induced drag ($e = 1.0$). Tapered wings approximate elliptical distribution.	Supermarine Spitfire (pure elliptical); modern airliners use trapezoidal tapered wings with aerodynamic washout.
Winglets / Sharklets	Vertical wingtip fences diffuse tip vortices, converting vortex energy into forward aerodynamic thrust.	Airbus sharklets / Boeing blended winglets reduce fuel burn by 3% to 5% in cruise.
Wing Washout (Twist)	Geometric wing twist (lower incidence at tip than root).	Ensures wing root stalls FIRST, preserving aileron roll control at high angles of attack.

[!] EXAM TRAP (AVIATIONEXAM) - Wake Turbulence & Heavy Jet Vortex Hazard

- When is Wake Turbulence Greatest?: When the generating aircraft is HEAVY, CLEAN (flaps/gear up), and SLOW! - Rotation / Touchdown Points: Vortices begin at nose-wheel rotation and end when nose-wheel touches down. - Avoidance Rule: Rotate BEFORE the preceding heavy aircraft's rotation point, and land BEYOND its touchdown point!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Aspect Ratio and Induced Drag**

Problem: Two wings have equal wing area $S = 20 \text{ m}^2$. Wing A has span 10 m; Wing B has span 20 m. Which has lower induced drag? Step 1: Compute Aspect Ratio $AR = b^2 / S$. For A: $AR = 10^2 / 20 = 5$. For B: $AR = 20^2 / 20 = 20$. Step 2: Induced drag coefficient C_{Di} is inversely proportional to AR. Wing B has 4 TIMES LESS induced drag than Wing A at the same C_L . Exam Rule: Longer wingspan = higher aspect ratio = lower induced drag.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: How does Induced Drag vary with aircraft airspeed in straight and level flight?**

- [A]: Directly proportional to airspeed
- [B]: Directly proportional to airspeed squared (V^2)
- [C]: Inversely proportional to airspeed SQUARED ($1 / V^2$)
- [D]: It remains constant

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. In level flight, as speed increases, C_L decreases by $1/V^2$. Because induced drag is proportional to C_L^2 , Induced Drag decreases with $1/V^2$, making it dominant at low speeds and tiny at high speeds.

Q2: What is the primary aerodynamic benefit of fitting winglets to modern jet transport aircraft?

- [A]: To increase maximum take-off mass
- [B]: To reduce Induced Drag by impeding spanwise airflow around wingtips and dissipating wingtip vortices
- [C]: To prevent deep stall
- [D]: To improve rudder authority

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Winglets act as vertical aerodynamic barriers that disrupt the high-to-low pressure spanwise bleed around wingtips, reducing vortex downwash and cutting induced drag by 3-5%.

Q3: Under what flight conditions does a transport aircraft generate the strongest, most hazardous wake turbulence?

- [A]: Fast, light, flaps fully extended
- [B]: HEAVY, CLEAN (flaps retracted), and SLOW
- [C]: Cruising at high Mach number
- [D]: In a steep dive

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Wake vortex strength is proportional to aircraft weight and inversely proportional to airspeed and wingspan. A heavy aircraft flying slowly in clean configuration generates maximum vortex circulation.

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 5: Total Drag, Minimum Drag Speed (V_{md}) & Ground Effect

Syllabus Source: CAE Oxford (pp. 145-184) | Pass Target: >= 90% AviationExam

1. PARASITE DRAG COMPONENTS: SKIN FRICTION, FORM & INTERFERENCE

Parasite drag includes all aerodynamic drag not associated with the generation of lift. It increases with the square of airspeed (D_p proportional to V^2):

- Skin Friction Drag: Result of viscous shearing stresses in the boundary layer. Minimized by keeping surfaces polished and maintaining laminar flow.
- Form Drag (Pressure Drag): Result of boundary layer separation creating a low-pressure wake behind bluff bodies. Minimized by streamlining and optimizing fineness ratio (length-to-diameter ratio ~3:1 to 4:1).
- Interference Drag: Result of turbulent mixing of boundary layers at structural junctions (wing root, fuselage, engine pylons). Minimized by aerodynamic fairings and root fillets.

2. TOTAL DRAG CURVE & MINIMUM DRAG SPEED (VMD)

Total drag is the sum of Induced Drag (D_i proportional to $1/V^2$) and Parasite Drag (D_p proportional to V^2):

- Vmd Equality Point: At Minimum Drag Speed (V_{md}), INDUCED DRAG EXACTLY EQUALS PARASITE DRAG: $D_i = D_p = 1/2$ Total Drag!
- Speed Stability Regimes: Speeds above V_{md} : Speed-stable (parasite drag dominates; slowing down reduces drag). Speeds below V_{md} : The 'Backside of the Drag Curve' / Reverse Command (induced drag dominates; flying slower REQUIRES MORE THRUST to maintain level flight!).
- Weight Variation: Increasing aircraft gross mass (W) shifts the entire total drag curve UP and to the RIGHT: V_{md} increases with the square root of mass (V_{md} proportional to $\sqrt{W}$); Minimum Total Drag increases directly proportional to mass (Total Drag proportional to W).

3. GROUND EFFECT AERODYNAMICS

When an aircraft flies within one wingspan distance above the ground ($h \leq b$, especially pronounced at $h \leq b/4$):

Aerodynamic Parameter	Physical Change Entering Ground Effect	Cockpit Symptoms & Pilot Action
Aerodynamic Parameter	Physical Change Entering Ground Effect	Cockpit Symptoms & Pilot Action
Induced Downwash & Vortices	Ground physically blocks vertical flow, suppressing wingtip vortices and reducing downwash angle.	Induced drag DROPS by up to 50%!
Effective Angle of Attack	Reduction of downwash rotates lift vector forward toward the vertical.	Lift INCREASES significantly at the same geometric pitch attitude.
Elevator Control Effectiveness	Reduced downwash over the tailplane decreases horizontal stabilizer download.	Generates an uncommanded NOSE-DOWN pitching moment on landing!
Airspeed Indicator Static Vent	Local static pressure at the vent increases slightly.	ASI and Altimeter read slightly LOWER than actual in ground effect.

[!] EXAM TRAP (AVIATIONEXAM) - Ground Effect Traps on Take-off and Landing

- LANDING (Floating): Due to the sudden reduction in induced drag, an aircraft entering flare with excess airspeed will FLOAT down the runway, risking runway overrun! - TAKE-OFF (Premature Liftoff Stall): An aircraft can lift off prematurely at a dangerously low speed inside ground effect. As it climbs out of ground effect (above $h > b/2$), induced drag suddenly increases and effective AOA drops, causing the aircraft to stall or settle back onto the runway!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Speed Instability & Ground Effect**

Rule 1 (Back Side of Drag Curve): If flying below V_{MD} on approach, do NOT pull back on the yoke to recover altitude if speed is decaying. Below V_{MD} , slower speed increases total drag. You must add thrust immediately. Rule 2 (Premature Liftoff Trap): In ground effect, induced drag drops by up to 50%. The aircraft may lift off below normal rotation speed V_r . Upon climbing out of ground effect (above half wingspan), induced drag surges, risking an immediate stall.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: At the Minimum Drag Speed (V_{md}) in straight and level flight, what is the exact mathematical relationship?

- [A]: Induced Drag is twice Parasite Drag
- [B]: Induced Drag EXACTLY EQUALS Parasite Drag ($D_i = D_p = 50\%$ of Total Drag)
- [C]: Induced Drag is zero
- [D]: Parasite Drag is zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At the minimum point of the total drag curve (V_{md}), the decreasing induced drag curve intersects the increasing parasite drag curve. Induced drag and parasite drag are exactly equal.

Q2: How does an increase in aircraft gross mass affect the Minimum Drag Speed (V_{md}) and minimum total drag?

- [A]: V_{md} decreases and drag increases
- [B]: Both V_{md} and minimum total drag INCREASE
- [C]: V_{md} increases but drag remains constant
- [D]: Neither changes

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Increasing weight requires higher lift. Higher lift requires higher speed to minimize drag (V_{md} increases proportional to $\sqrt{W}$) and produces higher total drag at that speed.

Q3: What occurs aerodynamically when an aircraft enters Ground Effect during landing flare?

- [A]: Induced drag increases and lift decreases
- [B]: Induced drag drops significantly, effective AOA increases, and the aircraft experiences an uncommanded nose-down pitching moment
- [C]: Skin friction drag triples

- [D]: The stall speed increases

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The proximity of the runway suppresses wingtip vortices and downwash, slashing induced drag, boosting lift (causing floating), and reducing tailplane downforce (causing a nose-down pitch).

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 6: Stalling Mechanics, Boundary Layer & Flow Separation

Syllabus Source: CAE Oxford (pp. 185-224) | Pass Target: >= 90% AviationExam

1. THE BOUNDARY LAYER: LAMINAR VS TURBULENT

The boundary layer is the microscopic fluid layer adjacent to the solid surface where viscous forces slow velocity from zero (surface no-slip) up to 99% of freestream:

Boundary Layer Characteristic	Laminar Boundary Layer	Turbulent Boundary Layer
Boundary Layer Characteristic	Laminar Boundary Layer	Turbulent Boundary Layer
Fluid Particle Motion	Smooth, parallel microscopic streamlines with zero cross-mixing.	Vortical, swirling eddies with intense kinetic momentum exchange.
Layer Thickness & Profile	Extremely thin; steep velocity gradient at wall.	Thicker; flatter velocity profile with energetic fluid near wall.
Skin Friction Drag	VERY LOW (ideal for laminar flow airfoils).	HIGHER (roughly double the friction drag of laminar).
Separation Resistance	POOR. Lacks kinetic energy; separates readily under mild adverse pressure gradient.	EXCELLENT. Kinetic mixing energizes lower layers, delaying flow separation against adverse pressure gradients!

- Transition Point: The point where the boundary layer transitions from laminar to turbulent. Moves FORWARD with: increasing airspeed/Reynolds number, increasing angle of attack, adverse pressure gradient, and surface roughness.

2. AERODYNAMIC STALLING MECHANICS

As angle of attack increases past the maximum lift coefficient (C_{L_max}), the adverse pressure gradient on the upper surface decelerates boundary layer air to a complete standstill, reversing flow direction and causing MASSIVE FLOW SEPARATION:

- Critical Angle of Attack (α_{crit}): The stalling angle of attack (typically 15° to 18°). An airfoil ALWAYS stalls at the EXACT SAME critical angle of attack, regardless of airspeed, gross mass, bank angle, or altitude!
- Stall Speed Formula: $V_S = \sqrt{(2 \times W) / (\rho \times S \times C_{L_max})}$.
- Factors Affecting Stall Speed: 1. Mass: V_S increases proportional to $\sqrt{W}$; 2. Load Factor (n): $V_S = V_{S0} \times \sqrt{n} = V_{S0} \times \sqrt{1 / \cos(\text{bank})}$; 3. Center of Gravity: Forward CG increases stall speed (larger download on tailplane increases effective wing loading); 4. Flaps: Flap extension increases C_{L_max} , lowering stall speed.

[!] EXAM TRAP (AVIATIONEXAM) - Accelerated Stall & Steep Turns Calculation Trap

- In a level coordinated turn, Load Factor $n = 1 / \cos(\text{Bank Angle } \Phi)$: - At 45° Bank: $n = 1.41 \text{ g} \rightarrow$ Stall Speed increases by $\sqrt{1.41} = +19\%$! - At 60° Bank: $n = 2.00 \text{ g} \rightarrow$ Stall Speed increases by $\sqrt{2.00} = 1.414 (+41.4\%)$! - If an aircraft stalls at 60 kt clean: At 60° bank it will stall at $60 \times 1.414 = 85 \text{ KNOTS}$!

3. WING PLANFORM STALL PROGRESSION PATTERNS

- Rectangular Wing: Stalls at WING ROOT first. Safest planform: root stall buffet provides natural pilot warning; ailerons at wingtips remain in unstalled air, retaining full roll control.
- Tapered / Swept Wing: Stalls at WINGTIP first. Dangerous: tip stall causes loss of aileron roll control and creates a severe, uncommanded NOSE-UP pitching moment (washout twist and stall strips are added to force root stall).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Accelerated Stall in Banked Turn

Problem: Normal stall speed in level flight is 60 kt ($V_{S0} = 60$ kt). What is the stall speed in a coordinated level turn at 60 degrees bank? Step 1: Compute load factor $n = 1 / \cos(60) = 1 / 0.5 = 2.0$ g. Step 2: Apply accelerated stall formula: $V_{S_turn} = V_{S0} \times \sqrt{n} = 60 \times \sqrt{2} = 60 \times 1.414 = 84.8$ kt. Memory Rule: In a 60-degree banked level turn, stall speed ALWAYS increases by 41% (multiplier 1.41).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: If an aircraft with a clean level-flight stall speed of 70 knots enters a coordinated 60° banked turn, what is the stall speed?

- [A]: 70 knots
- [B]: 85 knots
- [C]: 99 knots ($70 \times \sqrt{2}$)
- [D]: 140 knots

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Load factor at 60° bank is $n = 1 / \cos(60^\circ) = 2.0$ g. The stalling speed formula is $V_{S_turn} = V_{S0} \times \sqrt{n} = 70 \times \sqrt{2} = 70 \times 1.414 = 98.98$ (~99 knots).

Q2: Why does a TURBULENT boundary layer resist flow separation better than a LAMINAR boundary layer?

- [A]: Turbulent flow has zero viscosity
- [B]: Turbulent vortices mix high-energy air from the outer freestream down into the lower layer, energizing it to overcome adverse pressure gradients
- [C]: Turbulent flow is thinner
- [D]: Turbulent flow has lower skin friction

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The defining advantage of turbulent boundary layers is kinetic turbulent mixing. High-speed fluid from above is churned down near the wall, allowing the flow to remain attached longer against opposing adverse pressure gradients.

Q3: How does moving the aircraft Center of Gravity (CG) FORWARD affect the clean stall speed?

- [A]: Reduces stall speed
- [B]: INCREASES stall speed slightly
- [C]: Has zero effect on stall speed
- [D]: Causes an unrecoverable spin

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Forward CG increases the nose-down moment, requiring a greater downward aerodynamic force from the tailplane for trim. This extra downward load adds to the aircraft weight, increasing wing loading and raising stall speed.

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 7: High-Lift Devices (Flaps & Slats) & Contamination

Syllabus Source: CAE Oxford (pp. 225-264) | Pass Target: >= 90% AviationExam

1. TRAILING-EDGE FLAPS: CAMBER & AREA EXPANSION

Trailing-edge flaps increase maximum lift coefficient (C_{L_max}) to reduce take-off and landing speeds:

Flap Type	Mechanism of Operation & Boundary Layer Control	Aerodynamic Characteristics
Flap Type	Mechanism of Operation & Boundary Layer Control	Aerodynamic Characteristics
Plain Flap	Simple hinged trailing edge deflects downward, increasing camber.	Increases C_{L_max} moderately; moderate drag increase.
Split Flap	Deflects lower surface downward while upper surface remains fixed.	Large suction wake behind split plate; massive parasite drag increase; ideal dive brake.
Slotted Flap	Leaves an aerodynamic converging slot between wing and flap.	High-pressure lower air shoots through slot as high-velocity jet, energizing flap boundary layer and delaying flow separation.
Fowler Flap	Translates REARWARD on tracks before deflecting downward.	INCREASES BOTH CAMBER AND WING SURFACE AREA (S)! Achieves the highest C_{L_max} increase of any flap design! Modern airliners use multi-slotted Fowler flaps.

- Aerodynamic Effects of Flap Extension: 1. Increases C_{L_max} (lowers stall speed V_S); 2. Increases Drag (steepens approach descent angle); 3. Generates NOSE-DOWN pitching moment (due to rearward movement of center of pressure); 4. REDUCES the critical stalling angle of attack (α_{crit} decreases by $\sim 2^\circ$ to 4° !).

2. LEADING-EDGE HIGH-LIFT DEVICES: SLATS & SLOTS

Leading-edge slats allow an aircraft to operate at dramatically higher angles of attack without stalling:

- Slats & Slots Principle: A leading-edge slat opens a convergent nozzle slot that ducts high-energy air onto the upper surface. This delays flow separation at high angles of attack.
- α_{crit} Extension: Unlike trailing-edge flaps (which decrease α_{crit}), LEADING-EDGE SLATS EXTEND THE STALLING ANGLE OF ATTACK from $\sim 15^\circ$ up to 25° or even 30° ! Slats do not change zero-lift AOA.
- Krueger Flaps: Folding leading-edge flaps that rotate out from beneath the lower wing leading edge. Used near the engine nacelles on Boeing 747/777/737.

3. AIRFRAME CONTAMINATION: THE CLEAN AIRCRAFT CONCEPT

Frost, ice, and snow clinging to aerodynamic surfaces degrade flight safety catastrophically:

- Surface Roughness Effect: Even a thin frost layer with the roughness of medium sandpaper:
 1. Reduces maximum lift (C_{L_max}) by 25% to 30%!
 2. Increases stall speed by 15% to 20%!
 3. Increases drag by over 100%!
 4. Drastically reduces stalling angle of attack.

- Clean Aircraft Concept (EASA Mandate): Take-off is STRICTLY PROHIBITED unless all critical aerodynamic lifting surfaces, control surfaces, and engine intakes are completely free from frost, ice, snow, and slush contamination!

[!] EXAM TRAP (AVIATIONEXAM) - Flaps vs Slats Stalling Angle of Attack Traps

- Trailing-Edge Flaps: INCREASE C_{L_max} , but DECREASE the stalling angle of attack (α_{crit} drops by 2-4°)! - Leading-Edge Slats: INCREASE C_{L_max} , and DRAMATICALLY INCREASE the stalling angle of attack (α_{crit} increases from ~15° to 25°+)! - Frost / Ice on Wings: Reduces C_{L_max} AND reduces α_{crit} (wing can stall prematurely at take-off pitch attitude)!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Slats vs Flaps Operational Distinction

Exam Query: Which high-lift device increases C_{L_max} while reducing the stall angle of attack? Step 1: Check effect on critical angle. If it REDUCES critical angle -> It is a trailing edge FLAP. Step 2: If it INCREASES critical angle (delays stall to higher angles up to 25 degrees) -> It is a leading edge SLAT. Memory Rule: Slats = Super Angle (increases stall angle). Flaps = Fast Lift (reduces stall speed and stall angle).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What happens to the critical stalling angle of attack (α_{crit}) when TRAILING-EDGE FLAPS are extended?

- [A]: Stalling angle of attack increases
- [B]: Stalling angle of attack DECREASES (wing stalls at a lower AOA than clean)
- [C]: Stalling angle remains unchanged
- [D]: Stalling angle becomes 45°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Extending trailing-edge flaps increases camber, raising the upper surface adverse pressure gradient. This causes the boundary layer to separate earlier, reducing the critical stalling AOA by 2° to 4°.

Q2: What is the primary aerodynamic mechanism of a Fowler flap compared to a plain flap?

- [A]: It reduces parasite drag
- [B]: It extends rearward on tracks before deflecting, increasing both wing surface area (S) and camber, providing superior C_{L_max} increase
- [C]: It prevents Dutch roll
- [D]: It trims the rudder

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Fowler flaps translate aft on tracks to increase effective wing planform area before deflecting down to increase camber, yielding the highest C_{L_max} increase of any flap type.

Q3: How does a thin layer of frost on the wing upper surface affect take-off aerodynamic characteristics?

- [A]: It has zero effect

- [B]: It reduces maximum lift by up to 30%, increases stall speed significantly, and reduces the critical stalling angle of attack
- [C]: It lowers fuel consumption
- [D]: It acts as a laminar flow generator

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Frost acts like coarse sandpaper, tripping the laminar boundary layer prematurely into a thick, low-energy turbulent layer that separates prematurely, causing stall at normal take-off rotation angles.

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 8: Transonic Aerodynamics, Critical Mach & Shock Waves

Syllabus Source: CAE Oxford (pp. 265-304) | Pass Target: >= 90% AviationExam

1. MACH REGIMES & CRITICAL MACH NUMBER (MCRIT)

Compressibility effects dominate high-speed flight as airspeed approaches the speed of sound ($M = TAS / a$):

- Mach Flight Regimes: Subsonic: $M < 0.75$ (flow everywhere is subsonic); Transonic: $0.75 \leq M < 1.20$ (mixed subsonic and supersonic flow over airframe); Supersonic: $1.20 \leq M < 5.0$ (flow everywhere is supersonic); Hypersonic: $M \geq 5.0$.
- Critical Mach Number (M_{crit}): The free-stream Mach number at which airflow over the point of maximum curvature on the upper wing surface first accelerates to local Mach 1.0 (local sonic speed).
- Drag Divergence Mach Number (M_{dd}): The free-stream Mach number at which aerodynamic drag rises sharply (standard definition: where C_D increases by 0.0020 or 20 drag counts). Occurs slightly above M_{crit} (typically $M_{crit} + 0.06$ to 0.08).

2. NORMAL SHOCK WAVE PHYSICS & WAVE DRAG

When local airflow exceeds Mach 1.0 on the wing upper surface, it decelerates back to subsonic speed through a NORMAL SHOCK WAVE:

Fluid Property Across Normal Shock		
Fluid Property Across Normal Shock	Upstream Supersonic Flow (Ahead of Shock)	Downstream Subsonic Flow (Behind Shock)
Flow Velocity / Mach Number	Supersonic ($M_1 > 1.0$)	SUBSONIC ($M_2 < 1.0$). Discontinuous deceleration!
Static Pressure (p)	Lower ambient pressure	DISCONTINUOUS INSTANTANEOUS RISE (Compression step)!
Air Temperature & Density	Lower temperature and density	DISCONTINUOUS INSTANTANEOUS RISE!
Total Pressure & Usable Energy	Higher stagnation pressure	DECREASES SIGNIFICANTLY (Energy lost as heat/entropy)!

- Wave Drag: The massive drag rise caused by energy dissipation across the shock wave combined with boundary layer separation triggered by the shock's adverse pressure jump (Shock-Induced Separation / Mach Buffet).

3. TRANSONIC FLIGHT INSTABILITIES: MACH TUCK & MACH TRIM

As an aircraft accelerates beyond M_{crit} into the transonic region, it experiences severe trim and stability changes:

- Mach Tuck (Nose-Down Pitching): Caused by: 1. Rearward shift of the Center of Pressure from 25% chord toward 50% chord as supersonic flow expands aft; 2. Loss of downwash over the horizontal stabilizer due to shock-induced boundary layer separation at the wing root. Both effects create a dangerous, uncommanded NOSE-DOWN PITCH (Mach Tuck)!
- Mach Trim System: An automatic electrical trim system commanded by the Air Data Computer. As Mach number exceeds a preset threshold (e.g. $M 0.78$), the Mach trim actuator applies progressive UP-ELEVATOR / NOSE-UP STABILIZER TRIM to counter Mach tuck automatically without pilot input.

[!] EXAM TRAP (AVIATIONEXAM) - Airflow Velocity Across a Normal Shock Wave Rule

- CRITICAL EXAM TRAP: Airflow passing through a NORMAL shock wave ALWAYS leaves the shock at a SUBSONIC velocity ($M < 1.0$)! - Airflow passing through an OBLIQUE shock wave remains SUPERSONIC ($M > 1.0$)! - Static pressure, temperature, and density ALWAYS INCREASE across a shock wave, while total pressure ALWAYS DECREASES.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Normal Shock Wave & Mach Tuck**

Rule 1 (Normal Shock): Remember 'P-T-D rise, M-V-Ptot fall'. Across a normal shock wave, static Pressure, Temperature, and Density INCREASE; while Mach, Velocity, and Total Pressure DECREASE. Rule 2 (Mach Tuck): Above M_{crit} , shock-induced separation at wing root moves CP aft from 25% to 50% chord. The aircraft pitches down violently. The Mach Trim system automatically deflects the elevator nose-up to counter tuck.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the aerodynamic definition of the 'Critical Mach Number' (M_{crit})?**

- [A]: The Mach number where the aircraft enters a flat spin
- [B]: The free-stream Mach number at which airflow over any part of the aircraft first reaches local Mach 1.0
- [C]: The speed of sound at sea level
- [D]: Maximum operating Mach (M_{MO})

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. M_{crit} is the aircraft flight Mach number where the fastest local airflow accelerating over the wing upper surface first achieves sonic speed (local Mach 1.0).

Q2: What happens to flow velocity, static pressure, and total pressure as air passes through a NORMAL SHOCK WAVE?

- [A]: Velocity increases, static pressure decreases, total pressure increases
- [B]: Velocity drops from supersonic to SUBSONIC, static pressure RISES abruptly, and total pressure DECREASES
- [C]: All parameters remain constant
- [D]: Velocity remains supersonic

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A normal shock wave causes an irreversible adiabatic compression: flow velocity decelerates to subsonic ($< M 1.0$), static pressure steps up sharply, and total pressure drops due to entropy generation.

Q3: What causes 'Mach Tuck' in transonic jet transport aircraft?

- [A]: Flap extension at high speed
- [B]: Rearward movement of the wing Center of Pressure and reduction of tailplane downwash due to shock formation, inducing an uncommanded nose-down pitching moment
- [C]: Engine flameout
- [D]: Rudder deflection

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Shock wave development moves the center of lift rearward toward 50% chord and disrupts downwash onto the tailplane, producing an uncommanded nose-down diving tendency countered by Mach trim.

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 9: Transonic Design Features: Swept Wings & Area Rule

Syllabus Source: CAE Oxford (pp. 305-344) | Pass Target: >= 90% AviationExam

1. WING SWEEPBACK: THE EFFECTIVE VELOCITY VECTOR

Sweeping the wing rearward (sweep angle Λ , typically 25° to 35°) is the primary design technique to increase M_{crit} and M_{dd} :

- Velocity Decomposition: The oncoming freestream airspeed (V) is resolved into two orthogonal components: 1. Normal Component ($V_n = V \times \cos(\Lambda)$), perpendicular to the leading edge; 2. Spanwise Component ($V_s = V \times \sin(\Lambda)$), parallel to the leading edge.
- The Sweep Benefit: ONLY THE NORMAL COMPONENT (V_n) produces aerodynamic pressure changes and accelerates over the airfoil! The spanwise component produces only minor skin friction. Because $V_n = V \times \cos(\Lambda)$ is significantly smaller than V , the effective Mach number sensed by the wing is REDUCED: $M_{effective} = M \times \cos(\Lambda)$!
- Result on M_{crit} : Sweeping a wing increases the aircraft's Critical Mach Number: $M_{crit}(swept) = M_{crit}(straight) / \cos(\Lambda)$. Allows the aircraft to cruise at much higher speeds before shock waves form!

2. SWEPT WING AERODYNAMIC DISADVANTAGES & REMEDIES

While wing sweep raises cruising Mach number, it introduces severe low-speed and stability penalties:

Swept Wing Penalty / Problem	Aerodynamic Physical Cause	Engineering Solution & Cockpit Remedy
Swept Wing Penalty / Problem	Aerodynamic Physical Cause	Engineering Solution & Cockpit Remedy
Wingtip Stall Tendency	Spanwise airflow boundary layer drift thickens at tips, causing wingtips to stall FIRST.	Wing washout (twist), leading-edge stall fences, vortilons, and vortex generators.
Severe Pitch-Up at Stall	Tip stall occurs behind CG on swept wings; loss of tip lift shifts resultant lift FORWARD!	Stick Pusher system; low-mounted horizontal stabilizer to avoid deep stall wash.
Dutch Roll Instability	Strong dihedral effect of swept wings produces coupled yaw-roll oscillation.	Yaw Damper system (mandatory for flight dispatch under MEL).
Reduced $C_{L_{max}}$ & High Stall Speed	Swept wings have a flatter lift curve slope ($dC_L/d\alpha$ is lower).	Complex multi-slotted Fowler flaps and leading-edge slats mandatory for take-off/landing.

3. SUPERCRITICAL AIRFOILS & THE WHITCOMB AREA RULE

- Supercritical Airfoil: Features a FLATTENED upper surface (limits peak airflow acceleration, delaying and weakening shock waves), high aft camber (restores lift near the trailing edge), and a well-rounded leading edge. Allows higher cruising Mach numbers or thicker wings (saving structural weight).
- Whitcomb Area Rule: To minimize transonic wave drag, the cross-sectional area of the entire aircraft (fuselage + wings + engines) plotted along the longitudinal axis must change SMOOTHLY without abrupt steps. Achieved by 'pinching' the fuselage at the wing root (coke-bottle fuselage waist) or adding flap track canoe fairings / Küchemann carrots.

- Vortex Generators: Small, low-aspect-ratio vertical blades mounted on the wing upper surface at an angle of attack. Shed high-energy vortices that mix freestream energy into the boundary layer, preventing shock-induced flow separation.

[!] EXAM TRAP (AVIATIONEXAM) - Effective Mach Number on a Swept Wing Calculation

- Formula: $M_{\text{effective}} = M_{\text{free}} \times \cos(\text{Sweep Angle } \Lambda)$! - Example: An aircraft with 30° wing sweep flies at Mach 0.80. - The wing airfoil behaves as if it were flying at: $M_{\text{eff}} = 0.80 \times \cos(30^\circ) = 0.80 \times 0.866 = \text{MACH } 0.69$! - This delays shock wave formation by over 0.10 Mach!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Swept Wing Effective Mach Calculation

Problem: An airliner flies at Mach 0.80 with 30 degrees wing sweep ($\cos 30 = 0.866$). What Mach number does the airfoil aerodynamically experience perpendicular to the leading edge? Step 1: Formula: $M_{\text{effective}} = M_{\text{flight}} \times \cos(\text{Sweep})$. Step 2: $M_{\text{effective}} = 0.80 \times 0.866 = \text{Mach } 0.69$. Conclusion: The aircraft flies at Mach 0.80 while its wing profile experiences only Mach 0.69, delaying wave drag.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Why does sweeping an aircraft wing rearward increase the Critical Mach Number (M_{crit})?

- [A]: It increases wing aspect ratio
- [B]: It decomposes oncoming airflow, so that only the component PERPENDICULAR to the leading edge ($V \times \cos(\Lambda)$) determines aerodynamic pressure acceleration
- [C]: It eliminates induced drag
- [D]: It shifts the CG forward

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The chordwise pressure distribution on a swept wing depends solely on the velocity component normal to the leading edge ($V_n = V \cos(\text{sweep})$). Because V_n is lower than true airspeed, M_{crit} is delayed to a higher speed.

Q2: What dangerous aerodynamic stalling characteristic is inherently associated with a sweptback wing?

- [A]: Wing root stall with severe nose-down pitch
- [B]: WINGTIP STALL first, which causes loss of aileron roll control and an uncommanded violent NOSE-UP pitching moment
- [C]: Inability to stall
- [D]: Immediate flameout

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Swept wings suffer spanwise boundary layer flow that thickens the tip boundary layer, causing wingtips to stall first. Because swept wingtips are located well behind the CG, losing tip lift causes a strong pitch-up.

Q3: What design feature characterizes a 'Supercritical' airfoil?

- [A]: A highly cambered upper surface

- [B]: A flattened upper surface to weaken shock waves, combined with high aft camber to generate lift
- [C]: A sharp knife-like leading edge
- [D]: Negative sweep

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Supercritical airfoils feature a flattened upper surface that delays local supersonic acceleration and weakens shock waves, while deep underside curvature near the trailing edge restores required lift.

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 10: Static & Dynamic Stability Fundamentals

Syllabus Source: CAE Oxford (pp. 345-384) | Pass Target: >= 90% AviationExam

1. STATIC STABILITY: INITIAL TENDENCY UPON DISTURBANCE

Stability defines an aircraft's response to an aerodynamic disturbance from trimmed equilibrium:

- Positive Static Stability: The INITIAL tendency to return toward original trimmed equilibrium following disturbance (creates an opposing restoring moment: $dC_M / d\alpha < 0$).
- Neutral Static Stability: The initial tendency to remain in the displaced position with zero restoring or diverging force ($dC_M / d\alpha = 0$).
- Negative Static Stability (Static Instability): The initial tendency to diverge further away from trimmed equilibrium ($dC_M / d\alpha > 0$).

2. DYNAMIC STABILITY: BEHAVIOR OVER TIME

Dynamic stability describes the time history of the motion following an initial statically stable response. Dynamic stability CAN ONLY EXIST if the aircraft possesses positive static stability first!

Dynamic Stability Condition	Time-History Motion Characteristics	Physical Restoring & Damping Forces
Dynamic Stability Condition	Time-History Motion Characteristics	Physical Restoring & Damping Forces
Positive Dynamic Stability	Oscillations decrease in amplitude over time (damped motion) until steady equilibrium is restored.	Aerodynamic damping (e.g. tailplane pitch damping) dissipates kinetic energy into air.
Neutral Dynamic Stability	Oscillations continue indefinitely with CONSTANT amplitude (undamped sine wave).	Zero net damping force.
Negative Dynamic Stability	Oscillations amplify progressively over time with growing amplitude (divergent oscillation)!	Energy is continuously pumped into the oscillation, leading to structural divergence.

3. LONGITUDINAL DYNAMIC MODES: SPPO VS PHUGOID

Disturbing an aircraft longitudinally excites two distinct coupled pitch-speed oscillations:

- Short Period Pitch Oscillation (SPPO): High frequency (period typically 1 to 2 seconds), heavily damped. Consists of rapid pitch angle and Angle of Attack oscillations at virtually constant airspeed! If under-damped or if pilot attempts manual corrections out-of-phase, can trigger catastrophic Pilot Induced Oscillations (PIO)!
- Phugoid Motion (Long Period Mode): Low frequency (period typically 20 to 60+ seconds), weakly damped. Consists of a slow exchange between kinetic energy (airspeed) and potential energy (altitude) at VIRTUALLY CONSTANT ANGLE OF ATTACK! Aircraft climbs and slows down, then dives and speeds up. Easily controlled by pilot or autopilot altitude hold.

[!] EXAM TRAP (AVIATIONEXAM) - Short Period (SPPO) vs Phugoid Distinction Trap

- Short Period Pitch Oscillation: Variable AOA, CONSTANT airspeed! (High frequency, dangerous for PIO). - Phugoid Oscillation: CONSTANT AOA, variable airspeed and altitude! (Low frequency, easily damped by pilot). - Strict Rule: An aircraft CANNOT be dynamically stable unless it is statically stable first!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Static vs Dynamic Stability**

Exam Problem: A gust displaces the nose up by 5 degrees. The aircraft pitches down to recover, but overshoots and begins oscillations of 6, 8, and 10 degrees. Classify its stability. Step 1 (Initial Action): It initiated a return to equilibrium -> POSITIVE STATIC STABILITY. Step 2 (Time History): The oscillations increased in amplitude (6, 8, 10) -> DIVERGENT -> NEGATIVE DYNAMIC STABILITY. Conclusion: The aircraft is statically stable but dynamically unstable.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the defining characteristic of an aircraft exhibiting POSITIVE STATIC STABILITY?**

- [A]: It oscillates indefinitely at constant amplitude
- [B]: Its INITIAL tendency following an aerodynamic disturbance is to develop forces that restore it toward original trimmed equilibrium
- [C]: It requires no pilot controls
- [D]: It flies at supersonic speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Static stability describes purely the immediate initial tendency upon release from a disturbed displacement. Positive static stability means restoring moments oppose the disturbance.

Q2: In longitudinal dynamic flight modes, how do angle of attack and airspeed behave during a classic PHUGOID mode?

- [A]: Angle of attack oscillates rapidly while airspeed remains constant
- [B]: Angle of attack remains virtually CONSTANT while airspeed and altitude oscillate slowly
- [C]: Both parameters remain completely frozen
- [D]: Pitch damping is zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The phugoid mode is a long-period trade between kinetic and potential energy. Because pitch attitude tracks the flight path, the angle of attack remains essentially constant while speed and altitude cycle.

Q3: Can an aircraft possess positive dynamic stability if it has negative static stability?

- [A]: Yes, always
- [B]: NO, dynamic stability requires positive static stability as a prerequisite
- [C]: Only at high Mach numbers
- [D]: Only with flaps down

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. If an aircraft is statically unstable, it immediately diverges away from equilibrium, preventing any oscillatory motion from developing. Therefore, positive static stability is mandatory for dynamic stability.

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 11: Longitudinal Stability & CG Margins (Neutral Point)

Syllabus Source: CAE Oxford (pp. 385-424) | Pass Target: $\geq 90\%$ AviationExam

1. LONGITUDINAL EQUILIBRIUM: WING & TAILPLANE MOMENTS

Pitch stability (rotation about the lateral axis) is dominated by the interaction between the wing and horizontal tailplane:

- **Wing Pitching Moment:** Cambered wings generate a constant nose-down pitching moment about their Aerodynamic Center (M_{AC}). Center of Gravity (CG) is positioned forward of the wing center of lift, creating an additional nose-down couple.
- **Tailplane Download (Downforce):** To balance these nose-down couples in trimmed flight, the horizontal tailplane must generate a continuous DOWNWARD AERODYNAMIC FORCE (Tail Download)! This downward force reduces effective total lift and increases fuel burn, but provides pitch stability.
- **Longitudinal Dihedral:** The angle of incidence of the horizontal stabilizer is set SMALLER than the wing incidence (tailplane incidence $<$ wing incidence). When the aircraft pitches up, the percentage increase in tailplane AOA is greater than the wing, generating an upward restoring lift that pitches the nose back down!

2. THE NEUTRAL POINT (NP) & STATIC MARGIN

The Neutral Point is the theoretical aerodynamic center of the ENTIRE AIRCRAFT (wing + fuselage + tail):

- **Neutral Point (NP):** The unique CG position where the overall aircraft pitching moment does not change with angle of attack ($dC_M / d\alpha = 0$). At the Neutral Point, static pitch stability is EXACTLY ZERO (neutral).
- **Static Margin:** The physical distance between the Center of Gravity and the Neutral Point, expressed as a percentage of the Mean Aerodynamic Chord (MAC): $\text{Static Margin} = (x_{NP} - x_{CG}) / MAC$.
- **Stable CG Regime:** For positive longitudinal stability, the Center of Gravity MUST BE LOCATED FORWARD of the Neutral Point! Static Margin must be POSITIVE (typically 5% to 15% MAC in civil transports).

CG Location Condition	Stability & Stick Force per 'g'	Performance, Drag & Stall Speed
CG Location Condition	Stability & Stick Force per 'g'	Performance, Drag & Stall Speed
Forward CG Limit	MAXIMUM stability. Heavy stick forces; high stick force per g. Excellent recovery from stall.	High tailplane download required -> higher effective weight -> HIGHER STALL SPEED, higher drag, higher fuel burn. Limited by elevator authority in landing flare!
Aft CG Limit	REDUCED stability. Very light stick forces; low stick force per g. Highly sensitive controls.	Lower tailplane download -> lower trim drag, LOWER FUEL BURN, lower stall speed. Limited by minimum mandatory static margin!
CG Aft of Neutral Point	STATICALLY UNSTABLE! Any pitch disturbance causes divergent pitch runaway into deep stall or structural overload!	STRICTLY PROHIBITED in certified civil transport aircraft!

[!] EXAM TRAP (AVIATIONEXAM) - Forward CG vs Aft CG Flight Characteristics Trap

- FORWARD CG: More stable, heavier stick forces, higher stall speed, higher fuel consumption, flare authority limited! - AFT CG: Less stable, lighter stick forces (danger of overstressing airframe), lower stall speed, lower fuel consumption (higher cruise range)! - Aft CG Limit Rule: The aft CG limit is determined primarily by the minimum acceptable static longitudinal stability margin!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): CG Position Impact on Cruise & S**

Rule 'Aft = Aerodynamic efficiency, Forward = Forgiving stability': Step 1 (Aft CG): Shorter arm from weight to wing CP -> Less downward tail load needed -> Wing produces less total lift -> Lower induced drag -> LOWER fuel burn and HIGHER cruise speed. Step 2 (Forward CG): Long arm -> Large downward tail load -> Wing must produce extra lift -> Higher induced drag -> Higher fuel burn, but high static margin and easy stall recovery.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What happens to aircraft static longitudinal stability and stick forces as the Center of Gravity moves progressively aft?**

- [A]: Stability increases and stick forces increase
- [B]: Stability DECREASES and stick force per 'g' DECREASES (controls feel lighter and more sensitive)
- [C]: Stability remains unaffected
- [D]: The aircraft becomes uncontrollable immediately

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Moving CG aft shortens the distance to the Neutral Point (reducing Static Margin). This weakens restoring moments, reducing pitch stability and lowering stick force per g, making the aircraft more agile but sensitive.

Q2: Why does an aircraft cruising with a FORWARD Center of Gravity burn more fuel than with an AFT Center of Gravity?

- [A]: Engine thrust vector is misaligned
- [B]: Forward CG requires a larger downward tailplane force to balance the nose-down moment, increasing total lift required and induced trim drag
- [C]: Fuel cannot flow to engines
- [D]: Forward CG increases skin friction

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Forward CG creates a large nose-down moment, requiring a strong downward tailplane aerodynamic force. The wings must produce extra lift to support both aircraft weight and this tail download, increasing induced drag and fuel burn.

Q3: What aerodynamic position represents the absolute maximum aft theoretical limit for positive static longitudinal stability?

- [A]: The wing leading edge
- [B]: The NEUTRAL POINT of the aircraft
- [C]: The 25% MAC point

- [D]: The elevator hinge line

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. If the CG reaches the Neutral Point, the Static Margin becomes zero (neutral stability). Moving the CG aft of the Neutral Point results in negative static stability (pitch divergence).

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 12: Directional & Lateral Stability
(Dutch Roll & Spiral)

Syllabus Source: CAE Oxford (pp. 425-464) | Pass Target: >= 90% AviationExam

1. DIRECTIONAL (WEATHERCOCK) STABILITY

Directional stability is stability about the normal (vertical) axis following a sideslip angle (Beta):

- Weathercock Principle: When an aircraft sideslips, relative wind strikes the fuselage and vertical tail from the side. The vertical stabilizer (fin) generates a horizontal aerodynamic restoring force behind the CG, yawing the nose back into the relative wind.
- Contributions to Directional Stability: Vertical Fin (major positive stabilizing contributor); Rear Fuselage (stabilizing); Swept Wings (stabilizing: retreating wing has higher effective span and profile drag, pulling nose straight); Forward Fuselage and Nacelles (DESTABILIZING: generate adverse destabilizing yaw moments ahead of CG).

2. LATERAL (DIHEDRAL) STABILITY & CROSS-COUPLING

Lateral stability is stability about the longitudinal axis (roll) when perturbed into a sideslip:

- Geometric Dihedral (Gamma): Upward angle of wings from root to tip. When wings roll into a sideslip, the low wing meets relative airflow at a HIGHER effective angle of attack than the high wing, generating more lift and rolling the wings back level!
- Wing Sweepback Effect: Massive contributor to lateral stability! When a swept-wing aircraft sideslips, the wing into the wind (leading wing) has reduced effective sweep ($\Lambda - \beta$), increasing its normal velocity component ($V \cos(\Lambda - \beta)$) and lift, while the trailing wing loses lift. This produces a powerful restoring roll moment into the slip!
- High Wing (Anhedral vs Dihedral): High-mounted wings experience airflow blockage by the fuselage during sideslip, increasing high-wing lift (keel effect). Because swept high-wing air transports (C-17, Antonov) possess EXCESSIVE lateral stability, they use ANHEDRAL (negative dihedral / drooped wings) to restore balance!

3. DYNAMIC MODES: DUTCH ROLL VS SPIRAL DIVERGENCE

Dynamic Mode	Relative Stability Balance	Coupled Flight Motion & Operational Hazard
Dynamic Mode	Relative Stability Balance	Coupled Flight Motion & Operational Hazard
Dutch Roll	Excessive Lateral Stability (Dihedral/Sweep) combined with WEAK Directional Stability.	A coupled out-of-phase yaw and roll motion tracing a figure-8 corkscrew trajectory. Annoying to passengers; potentially divergent at high altitudes. MANDATORY REMEDY: Yaw Damper!

Spiral Divergence	STRONG Directional Stability combined with WEAK Lateral Stability (Dihedral).	Aircraft sideslips into turn; strong fin weathercocks nose into turn, while weak dihedral fails to level wings. Bank angle and rate of descent steepen progressively into a high-speed spiral dive. Non-oscillatory; slow divergence; easily corrected by pilot.
--------------------------	---	--

[!] EXAM TRAP (AVIATIONEXAM) - Dutch Roll vs Spiral Stability Trade-off

- Designing for high lateral stability suppresses spiral dive, BUT CREATES DUTCH ROLL! - Designing for high directional stability suppresses Dutch roll, BUT CREATES SPIRAL DIVERGENCE! - Modern transport jets accept mild spiral divergence (easily controlled by autopilot) and eliminate Dutch roll using automatic YAW DAMPERS.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Dutch Roll vs Spiral Divergence

Rule 1 (Dutch Roll): Lateral Stability > Directional Stability -> Dihedral/sweep rights wings too aggressively compared to weak rudder/fin damping, causing coupled out-of-phase yawing and rolling. Countered with Yaw Damper. Rule 2 (Spiral Divergence): Directional Stability > Lateral Stability -> Strong vertical fin aligns nose with wind before dihedral levels wings. Bank angle tightens into a descending spiral. Exam Rule: Dutch Roll = Strong Lateral; Spiral = Strong Directional.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What stability combination causes an aircraft to be susceptible to DUTCH ROLL?

- [A]: Strong directional stability and weak lateral stability
- [B]: Strong LATERAL stability (dihedral effect) combined with relatively WEAK DIRECTIONAL stability
- [C]: Forward CG
- [D]: Zero sweepback

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Dutch roll occurs when roll restoring forces (dihedral/sweep) are much stronger than directional yaw restoring forces. The aircraft over-corrects in roll before yaw can damp out, resulting in a corkscrew oscillation.

Q2: Why do many military cargo aircraft with swept, high-mounted wings incorporate ANHEDRAL (drooped wingtips)?

- [A]: To increase cruise speed
- [B]: To reduce excessive lateral stability created by the combination of high wing position and wing sweep, which would otherwise induce violent Dutch roll
- [C]: To assist engine maintenance
- [D]: To increase take-off lift

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Both swept wings and high-wing placement generate strong lateral stability (effective dihedral). Left unchecked, this makes the aircraft violently prone to Dutch roll. Anhedral is added to reduce roll-due-to-sideslip to a safe balance.

Q3: What is the primary operational function of an aircraft Yaw Damper system?

- [A]: To steer the nosewheel on landing
- [B]: To detect and immediately suppress Dutch Roll oscillations by commanding automatic rudder deflections
- [C]: To prevent stalls
- [D]: To trim the ailerons

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A yaw damper uses rate gyros to sense yaw rate and commands small, fast opposing rudder deflections via servo actuators, damping out Dutch roll without pilot pedal movement.

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 13: Flight Controls, Control Tabs & Aerodynamic Balance

Syllabus Source: CAE Oxford (pp. 465-504) | Pass Target: >= 90% AviationExam

1. PRIMARY CONTROLS & ADVERSE YAW MITIGATION

Primary flight controls govern rotation about the aircraft's three axes (Ailerons: Roll; Elevator: Pitch; Rudder: Yaw):

- **Adverse Yaw Phenomenon:** When ailerons deflect to initiate a roll, the downward-deflected aileron on the rising wing produces more lift AND MORE INDUCED DRAG than the upward-deflected aileron on the falling wing. This differential drag yaws the nose in the direction OPPOSITE to the commanded bank!
- **Differential Ailerons:** The up-going aileron deflects through a substantially LARGER angle (e.g. 25° up) than the down-going aileron (e.g. 10° down). The large up-deflection generates extra parasite form drag on the inside wing, counteracting adverse yaw.
- **Frise Ailerons:** The aileron hinge is set back from its leading edge. When deflected UP, the lower lip of the aileron protrudes below the wing lower surface into the slipstream, creating parasite drag on the down-going wing to balance adverse yaw.
- **Roll Spoilers:** Spoilers deflect upward ONLY on the down-going wing, destroying lift and adding large parasite drag on the inside of the turn, completely eliminating adverse yaw at high speeds.

2. CONTROL TABS & AERODYNAMIC BALANCING

Aerodynamic balancing reduces the physical hinge moment (stick force) required from the pilot:

Control Tab / Balance Device	Mechanical Deflection Direction	Aerodynamic Function & Stick Force Effect
Control Tab / Balance Device	Mechanical Deflection Direction	Aerodynamic Function & Stick Force Effect
Trim Tab (Controllable)	Deflects OPPOSITE to primary control surface (moved by cockpit trim wheel).	Holds control surface in deflected position with ZERO stick force from pilot. Trims aircraft for hands-off flight.
Balance Tab	Mechanically geared to deflect OPPOSITE to primary surface automatically.	Reduces pilot control stick forces by using tab aerodynamic lift to help deflect main surface.
Anti-Balance Tab	Mechanically geared to deflect in the SAME DIRECTION as primary surface.	INCREASES control feel and stick forces! Mandatory on all-flying stabilators (Piper Cherokee) to prevent pilot over-controlling and overstressing airframe.
Servo Tab	Directly connected to pilot cockpit controls; main surface is free-floating.	Pilot moves tab only; tab aerodynamic force pushes the entire large control surface into position (BAe 146, DC-9).
Spring Tab	Incorporates a torsion spring in the linkage.	At low speed, spring acts as rigid rod (direct manual control). At high dynamic pressure, spring compresses, allowing tab to assist pilot as a servo tab.

[!] EXAM TRAP (AVIATIONEXAM) - Balance Tab vs Anti-Balance Tab Direction Trap

- Balance Tab: Moves OPPOSITE to control surface -> REDUCES control stick forces! -
 Anti-Balance Tab: Moves in SAME DIRECTION as control surface -> INCREASES control stick forces and artificial feel! - Trim Tab: Cockpit trim nose-down commands trim tab UP, which aerodynamically forces the elevator DOWN.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Flight Control Tabs Deflection Sequence**

Problem: You trim elevator nose-up (primary surface UP). Which way does the trim tab deflect? Step 1: A conventional trim tab generates aerodynamic force to hold the trailing edge of the primary control. Step 2: To keep the elevator UP, the tab must generate a downward aerodynamic force -> Tab deflects DOWN. Exam Rule: Trim tab and Balance tab move in the OPPOSITE direction of the control surface. Anti-balance tab moves in the SAME direction.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: How do DIFFERENTIAL AILERONS reduce adverse yaw during a roll maneuver?**

- [A]: By disconnecting the rudder
- [B]: The up-going aileron deflects through a GREATER angle than the down-going aileron, producing extra parasite drag on the inside wing to counter adverse induced drag
- [C]: By lowering the flaps
- [D]: By moving both ailerons downward together

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The up-aileron deflects significantly farther than the down-aileron, creating deliberate form drag on the descending wing to counteract the adverse induced drag of the rising wing.

Q2: In what direction does an ANTI-BALANCE TAB move relative to the main control surface, and what is its purpose?

- [A]: In the opposite direction, to reduce control forces to zero
- [B]: In the SAME DIRECTION as the control surface, to INCREASE stick forces and provide artificial aerodynamic feel on an all-flying tail
- [C]: It remains locked in cruise
- [D]: It trims the rudder only

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. An anti-balance tab deflects in the same direction as the surface, opposing pilot input and increasing stick force per g, preventing over-controlling on light all-flying stabilizers.

Q3: When the pilot trims for NOSE-UP pitch using a conventional elevator trim tab, in which direction does the trim tab deflect?

- [A]: The trim tab deflects UP
- [B]: The trim tab deflects DOWNWARDS, creating an upward aerodynamic force that pushes and holds the elevator UP
- [C]: The tab does not move

- [D]: The tab moves laterally

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. To hold the elevator up for nose-up trim, the trim tab at the elevator trailing edge must deflect DOWN, generating upward aerodynamic lift that forces the elevator trailing edge up.

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 14: Flight Mechanics, V-n Diagram & Asymmetric Flight

Syllabus Source: CAE Oxford (pp. 505-548) | Pass Target: $\geq 90\%$ AviationExam

1. COORDINATED TURNS: LOAD FACTOR, RADIUS & RATE

In a level coordinated turn, aerodynamic lift is tilted inward at bank angle Φ , providing the horizontal centripetal force ($F_c = L \times \sin(\Phi)$):

- Load Factor (n): Vertical equilibrium requires $L \times \cos(\Phi) = \text{Weight (W)}$. Therefore: Load Factor $n = \text{Lift} / \text{Weight} = 1 / \cos(\Phi)$. At 60° bank: $n = 1 / \cos(60^\circ) = 2.0 \text{ g}$!
- Turn Radius Formula: Radius (R) = $V^2 / (g \times \tan(\Phi))$. Turn radius depends EXCLUSIVELY on True Airspeed and Bank Angle! For a given bank angle, radius increases with speed squared (doubling TAS quadruples turn radius!).
- Rate of Turn (ROT): $\text{ROT} = (g \times \tan(\Phi)) / V$. Rate 1 Turn = $3^\circ/\text{second}$ (360° turn in 2 minutes). Rule of thumb: Bank Angle for Rate 1 = $(\text{TAS} / 10) + 7$ (e.g. at 140 kt: $14 + 7 = 21^\circ$ bank).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Maneuvering Speed VA & Load F

Golden Rule of VA in Exams: Step 1 (Below VA): Full abrupt control deflection will cause an AERODYNAMIC STALL before exceeding limit load factor ($+2.5\text{g}$ transport / $+3.8\text{g}$ normal). The wing stalls and relieves load, protecting structure. Step 2 (Above VA): The wing has sufficient lift available to exceed limit load factor before stalling. Full abrupt deflection will cause STRUCTURAL DAMAGE or catastrophic failure. Formula: $V_A = V_{s1} \times \sqrt{n_{\text{limit}}}$.

2. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Load Factor and Turn Stall Speed in a Steep Turn

SCENARIO (Fundamental Flight Mechanics Exam Drill): An aircraft has a level, unaccelerated 1g stall speed $V_s = 60$ knots: - The pilot executes a level coordinated steep turn at 60° bank angle ($\Phi = 60^\circ$) QUESTION: What is the load factor (n), and what is the new accelerated stall speed (V_{st}) in the turn? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Load Factor (n): - Formula: $n = 1 / \cos(\text{Bank Angle})$
 - Bank Angle = $60^\circ \rightarrow \cos(60^\circ) = 0.50$ exactly! - Load Factor $n = 1 / 0.50 = 2.0 \text{ g}$!
 - (Both the aircraft structure and pilot experience twice their normal weight!). Step 2: Calculate the Accelerated Turn Stall Speed (V_{st}): - Master Formula: $V_{st} = V_s \times \sqrt{n}$ - $V_s = 60$ knots - $\sqrt{n} = \sqrt{2.0} = 1.414$ - $V_{st} = 60 \text{ kt} \times 1.414 = 84.8$ knots (~ 85 knots)! FINAL ANSWER: Load Factor = 2.0 g . Turn Stall Speed = 85 knots (a 41.4% increase in stall speed!). If airspeed drops below 85 kt during a 60° bank turn, the aircraft will stall immediately, even though it was flying safely above its 60 kt straight stall speed!

3. THE V-N DIAGRAM (MANEUVERING FLIGHT ENVELOPE)

The V-n diagram defines the structural and aerodynamic boundaries within which an aircraft may be safely operated:

Speed / Boundary on V-n Diagram	Regulatory Definition (CS-25 / CS-23)	Structural & Aerodynamic Limit Meaning
Speed / Boundary on V-n Diagram	Regulatory Definition (CS-25 / CS-23)	Structural & Aerodynamic Limit Meaning
Curved Left Boundary (Stall Line)	$n = C_{L_max} \times (1/2 \rho V^2) / (W/S)$.	Aerodynamic stall limit. Left of this line, the aircraft cannot generate enough lift to fly.
Maneuvering Speed (V_A)	Intersection of positive stall curve and limit load factor.	The MAXIMUM speed at which FULL, ABRUPT aerodynamic control deflection can be applied without exceeding structural limit load! (Aircraft stalls before structural failure).
Positive Limit Load Factor	+2.5 g (Transport CS-25); +3.8 g (Normal CS-23).	Maximum certified load factor without permanent structural deformation.
Ultimate Load Factor	1.5 x Limit Load Factor (+3.75 g for CS-25).	Airframe must withstand ultimate load for AT LEAST 3 SECONDS without catastrophic failure!
Design Dive Speed (V_D / M_D)	Maximum structural speed envelope limit.	Highest speed demonstrated in dive tests; margins above V _{MO} / M _{MO} .

4. MULTI-ENGINE ASYMMETRIC FLIGHT & V_{MCA}

When an engine fails on a multi-engine aircraft, asymmetric thrust creates severe yaw and roll toward the dead engine:

- The Critical Engine: On aircraft with conventional clockwise-rotating propellers, failure of the LEFT ENGINE is critical because the right engine's center of thrust acts farther from the fuselage centerline due to: 1. P-Factor (Asymmetric Blade Effect); 2. Accelerated Slipstream; 3. Torque Reaction; 4. Gyroscopic Precession.
- Minimum Control Speed Airborne (V_{MCA}): The minimum calibrated airspeed at which directional and lateral control can be safely maintained with the critical engine inoperative, maximum take-off power on the operating engine, and up to 5° OF BANK HELD INTO THE OPERATING ENGINE! If airspeed drops below V_{MCA}, rudder authority is lost and catastrophic uncontrollable roll/spin occurs!

[!] EXAM TRAP (AVIATIONEXAM) - V_{MCA} Bank Angle & Weight Dependencies

- 5° Bank Rule: Banking up to 5° TOWARDS THE LIVE ENGINE uses horizontal lift to assist the rudder, REDUCING V_{MCA} significantly! - Wings Level with Ball Centered: Increases V_{MCA} by 15-20 knots! Highly dangerous! - Weight Effect: Heavier aircraft has LOWER V_{MCA} (horizontal component of heavier lift vector helps counter yaw)! - Density Altitude: Higher altitude reduces engine thrust, which REDUCES yaw moment and LOWERS V_{MCA}!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the aerodynamic definition and safety significance of the 'Maneuvering Speed' (V_A)?

- [A]: The maximum speed for extending flaps
- [B]: The maximum speed at which full, abrupt deflection of flight controls can be made without risk of exceeding structural limit load factors, because the aircraft will stall aerodynamically before breaking
- [C]: The minimum drag speed

- [D]: The best rate of climb speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At speeds at or below V_A , the wing will stall before reaching the structural limit load factor (+2.5g on transport, +3.8g on utility), preventing structural overload during abrupt maneuvers.

Q2: Under CS-25 regulations, what is the 'Ultimate Load Factor' requirement for transport category aircraft?

- [A]: Exactly equal to the limit load
- [B]: 1.5 times the Limit Load Factor, which the structure must withstand for at least 3 seconds without collapsing
- [C]: 9.0 g
- [D]: 4.0 g

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Ultimate load incorporates a safety factor of 1.50 over the limit load (+2.5g x 1.5 = +3.75g). The airframe must sustain ultimate load for at least 3 seconds without structural collapse.

Q3: Why is the LEFT engine considered the 'critical engine' on conventional twin-engine piston aircraft with clockwise rotating propellers?

- [A]: Left engine has lower horsepower
- [B]: The right engine's center of thrust acts at a greater lateral moment arm from the fuselage centerline due to P-factor (asymmetric blade loading), generating more severe yaw if the left engine fails
- [C]: Left engine drives the alternator
- [D]: Propeller wash affects the cockpit

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Due to P-factor at high angles of attack, the descending blade produces more thrust. For clockwise propellers, this shifts the right engine's thrust line farther right, creating a larger yaw moment than the left engine.